

KING'S LAND O'LAKES, WI, USA

WMO: 723123

Lat: 46.150N

Lon: 89.217W

Elev: 1704

StdP: 13.81

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 14886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.7	-12.9	-26.1	1.4	-19.5	-18.6	2.1	-10.6	18.2	17.5	16.7	16.9	1.7	260	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	85.5	67.9	82.2	66.1	79.7	64.6	71.3	79.9	69.5	76.8	68.0	75.3	7.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.7	112.5	74.6	67.2	106.5	73.0	65.5	100.2	71.6	36.2	80.1	34.7	77.1	33.3	74.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.2	15.5	12.9	DB	-28.1	90.3	3.7	2.4	-30.8	92.0	-32.9	93.4	-35.0	94.8	-37.7	96.6	(4)
			WB	-28.4	74.7	3.6	2.1	-31.0	76.3	-33.1	77.5	-35.1	78.7	-37.7	80.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	12.9	13.2	25.9	37.9	51.6	60.6	64.8	62.0	55.3	43.2	30.3	18.1
	DBStd	21.05	12.94	12.60	13.27	10.01	9.35	6.96	6.33	6.34	8.58	8.94	10.20	11.48
	HDD50	5288	1149	1032	753	377	93	6	0	1	42	254	592	989
	HDD65	9385	1614	1452	1211	812	429	166	84	136	311	676	1040	1454
	CDD50	1567	0	0	7	15	141	323	460	374	200	45	2	0
	CDD65	190	0	0	0	0	12	33	79	44	20	2	0	0
	CDH74	2418	0	0	2	6	212	474	1006	501	200	17	0	0
	CDH80	522	0	0	0	0	51	95	254	90	32	0	0	0
Wind	WSAvg	5.4	5.7	5.8	5.8	6.4	6.1	5.2	4.6	4.2	4.7	5.5	5.7	5.4
Precipitation	PrecAvg	31.7	1.4	1.0	1.9	2.6	3.3	3.5	3.9	3.1	3.6	3.6	2.0	1.7
	PrecMax	38.3	3.6	2.1	4.2	5.0	6.2	6.7	8.4	6.4	7.6	6.8	4.8	2.7
	PrecMin	26.4	0.5	0.0	0.3	0.7	1.3	1.9	1.2	1.2	0.9	1.4	0.4	0.4
	PrecStd	4.1	0.8	0.6	1.1	1.3	1.2	1.5	1.6	1.2	1.4	1.5	1.0	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	50.0	69.8	73.8	85.8	87.6	89.7	86.1	84.0	76.7	61.3	44.4
		MCWB	36.7	41.3	57.1	55.9	64.0	69.7	71.1	68.1	68.4	59.8	51.8	39.8
	2%	DB	36.9	40.7	58.0	67.2	80.4	82.3	86.1	82.3	79.4	70.1	54.8	39.0
		MCWB	34.3	35.8	49.1	49.8	61.5	65.4	69.1	67.3	65.9	59.4	48.7	36.1
	5%	DB	33.7	34.4	51.6	62.1	74.8	79.4	82.4	79.4	75.8	64.4	49.6	35.2
		MCWB	32.5	31.4	45.3	47.2	58.6	63.0	66.8	65.2	63.7	56.6	44.1	33.8
	10%	DB	31.5	31.1	45.2	56.7	69.9	76.5	80.0	76.6	71.7	58.9	45.1	32.9
		MCWB	30.3	28.8	38.3	45.2	56.0	61.6	65.5	64.0	62.0	52.1	41.1	32.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	42.3	58.9	58.4	68.7	72.5	75.3	72.6	71.0	68.0	54.8	42.4
		MCDB	40.8	49.0	68.6	68.3	79.9	81.6	84.0	80.4	78.3	70.6	58.7	42.7
	2%	WB	34.7	35.9	52.7	53.8	65.3	69.0	71.8	70.4	68.6	61.3	49.4	36.6
		MCDB	36.6	39.2	55.8	62.5	72.8	77.9	81.1	77.1	74.8	67.7	54.2	38.0
	5%	WB	32.4	31.7	43.8	49.4	62.3	66.8	69.8	68.6	66.9	56.8	44.3	33.9
		MCDB	33.4	34.5	50.9	59.4	70.3	74.8	77.9	75.1	72.9	63.8	48.0	35.1
	10%	WB	29.6	28.7	38.9	46.0	59.1	64.5	68.0	66.8	64.2	52.8	41.3	32.1
		MCDB	30.9	30.5	44.0	55.9	67.3	71.3	76.0	73.6	69.9	57.9	44.8	33.1
Mean Daily Temperature Range	5% DB	MDBR	16.9	21.6	23.4	22.4	25.0	24.7	25.3	24.6	23.6	18.1	14.6	14.3
		MCDBR	18.5	22.5	26.9	33.9	31.5	29.3	28.3	27.4	27.7	27.0	23.0	13.7
		MCWBR	15.8	17.4	17.0	18.4	14.5	14.1	13.3	13.9	14.7	16.1	15.7	11.4
	5% WB	MCDBR	17.1	18.8	23.2	28.8	24.2	24.5	23.9	21.9	21.7	23.0	20.4	12.3
	MCWBR	15.0	15.0	15.7	17.2	12.1	13.5	12.7	12.3	13.2	15.4	15.1	10.8	
Clear-Sky Solar Irradiance	taub	0.237	0.264	0.305	0.327	0.359	0.360	0.361	0.357	0.339	0.306	0.280	0.249	
	taud	2.402	2.320	2.350	2.442	2.393	2.409	2.429	2.441	2.493	2.575	2.545	2.439	
	Ebn at Noon	283	292	293	296	288	287	285	282	279	276	264	265	
	Edh at Noon	22	29	33	34	36	36	35	33	29	23	19	19	
All-Sky Solar Radiation	RadAvg	426	750	1157	1443	1708	1841	1905	1601	1224	744	424	306	
	RadStd	39	83	107	146	141	141	93	100	74	84	49	34	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page